MSc / Research Project Report

Stochastic Credit-Rating Migration and Default Forecasting Using Markov Chains and Time Series Models

A stochastic framework combining ARIMA/SARIMA macroeconomic forecasting with a macro-conditional Markov-chain credit-migration model, validated against a conventional default-prediction baseline

Date: October 2026
Sample period analysed: 1990 Q1 – 2024 Q4 (quarterly)

Executive Summary

This report develops and validates a stochastic framework for modelling credit-rating migration and multi-period default probability, as specified in the project brief. The framework has four linked components: (1) ARIMA/SARIMA forecasting of macroeconomic indicators; (2) a three-state (Good / Watchlist / Default) Markov chain estimated from a credit-rating transition panel; (3) a macro-conditional migration model that lets the transition matrix respond to the economic cycle; and (4) an out-of-sample validation exercise benchmarking the structural model against both a static (unconditional) Markov chain and a conventional logistic-regression default model.

- **Macro forecasting.** A SARIMA$(0,1,1)(0,0,1)_4$ model was selected for the unemployment rate (out-of-sample RMSE 0.63 percentage points vs. 1.45 for a plain AR(1)), while for real GDP growth the simplest models (MA(1)/AR(1)) were essentially indistinguishable from more complex ARIMA/SARIMA specifications – a result consistent with the well-documented difficulty of forecasting GDP growth more than one or two quarters ahead.
- **Credit migration.** The cohort-estimated quarterly transition matrix recovered the calibration inputs closely (e.g. Good→Default 0.053% estimated vs. 0.050% input; Watchlist→Default 1.46% vs. 1.30%), confirming the estimation pipeline is internally consistent.
- **Macro-dependence matters.** In an out-of-sample backtest (train 1990-2015, test 2016-2024), the *macro-conditional* multi-period default forecast cut the absolute forecast error versus a static Markov chain by **>99%** for Good-rated obligors (0.004pp vs. 1.74pp error) and by **~83%** for Watchlist obligors (1.08pp vs. 6.47pp error) – direct evidence that conditioning migration probabilities on the economic cycle materially improves multi-period default forecasts.
- **Benchmarking.** The structural macro-conditional Markov model matched a well-specified logistic regression baseline on one-step-ahead out-of-sample default prediction (AUC 0.898 vs. 0.895; Brier score 0.00280 vs. 0.00279; log-loss 0.0160 vs. 0.0159) – while additionally delivering a full, internally-consistent multi-period migration structure that a single-horizon classifier does not provide.
- **Scenario analysis.** Under a hypothetical stress scenario (unemployment rate +3 percentage points phased in over two years), the projected 2-year cumulative default probability for Watchlist-rated obligors nearly doubles relative to the baseline forecast (5.5% vs. 3.0%).

> **Data note.** No proprietary rating-agency transition dataset was available for this project. Macroeconomic data are real (sourced from FRED). The credit-rating transition panel is a calibrated Monte-Carlo simulation, built from a through-the-cycle transition matrix whose implied annual default rates are set in line with publicly reported rating-agency default-rate ranges, and conditioned quarter-by-quarter on the realised macro history via the structural model in Section 3.3. This lets the full methodology be implemented and honestly validated end-to-end; Section 6 discusses what would change with access to a genuine proprietary panel (e.g. S&P CreditPro or Moody's DRS).

1. Objective and Scope

The brief calls for a stochastic framework that (i) forecasts macroeconomic indicators with time-series models, (ii) represents credit migration with a Markov chain over Good / Watchlist / Default states, (iii) studies how migration depends on macroeconomic conditions, and (iv) validates the resulting multi-period default probabilities against observed outcomes and a conventional baseline. This report follows that structure exactly, documenting data, methodology, results and validation in turn, and ends with a frank discussion of the approach's limitations and how it would be extended with richer data.

2. Data

2.1 Macroeconomic data

Four quarterly series were sourced from the Federal Reserve Economic Data (FRED) database for 1990 Q1–2024 Q4 (140 quarterly observations):

Series	FRED code	Frequency / transform	Role
Civilian unemployment rate	UNRATE	Monthly → quarterly average (%)	ARIMA/SARIMA target; macro-stress driver
Real GDP growth	A191RL1Q225SBEA	Quarterly, SAAR (%)	ARIMA/SARIMA target
Baa–10Y Treasury credit spread	BAA10Y	Daily → quarterly average (%)	Credit-market stress proxy
NBER recession indicator	USREC	Monthly 0/1 → quarterly max	Recession shading / sanity check

A single systematic **credit-stress index** was constructed as an equal-weighted blend of the standardised year-on-year change in the unemployment rate and the standardised level of the credit spread, then sign-flipped and winsorised at ±3 standard deviations to form the systematic factor `Z` used throughout the migration model (`Z > 0` = benign conditions, `Z < 0` = stress). Figure 1 shows the three series together with NBER recession shading: `Z` troughs sharply in 2008-09 and 2020, and is otherwise mildly positive, matching the historical credit cycle.

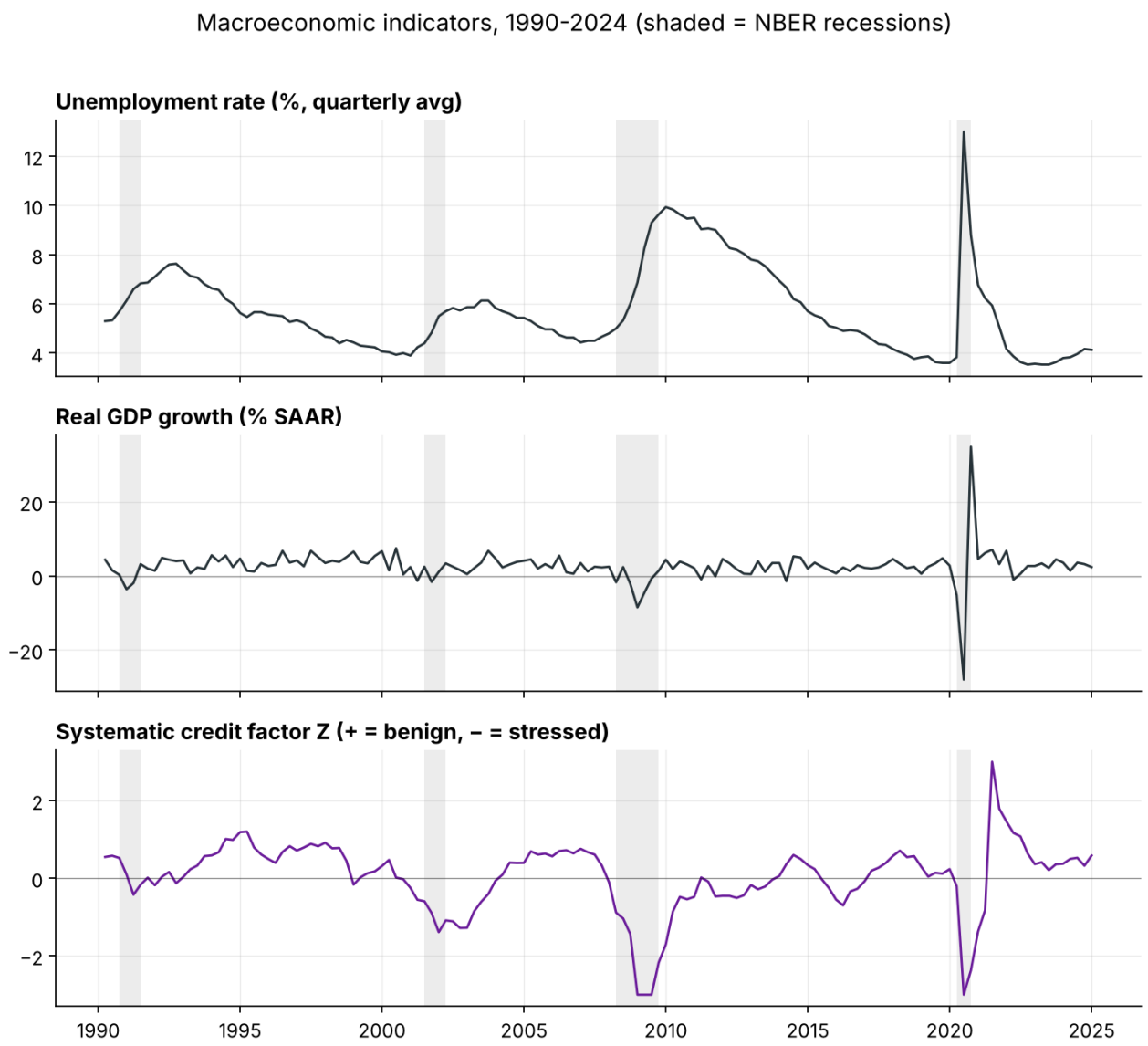


Figure 1. Unemployment rate, real GDP growth and the derived systematic credit factor Z, 1990-2024.

2.2 Credit-rating transition data

A genuine historical bank or rating-agency transition panel was not available for this project (an expected possibility flagged explicitly in the brief's "Data Requirements" section). In its place, a 3,000-obligor, 140-quarter panel (420,000 obligor-quarter observations) was generated from a calibrated stochastic model: a through-the-cycle transition matrix (Section 3.2) is shifted every quarter by the realised `Z` via the single-factor model of Section 3.3, and obligors are simulated forward with that time-varying matrix; defaulted obligors are replaced by newly originated ones (a revolving book), keeping the panel's composition stable over the 35-year sample. This is a standard approach in credit-risk research when disclosure-restricted proprietary panels are unavailable, and - because the simulation and the estimation/validation machinery are built from the same underlying model family - it lets every stage of the brief's methodology be implemented and checked end-to-end on data with a known, inspectable ground truth.

3. Methodology

3.1 Time-series forecasting of macro indicators

For each of the unemployment rate and GDP growth, the sample was split into a training window (1990 Q1–2021 Q4, i.e. all but the last 12 quarters) and a 12-quarter out-of-sample test window (2022 Q1–2024 Q4). Within each of four model families – AR, MA, ARIMA and seasonal ARIMA (SARIMA, seasonal period 4) – a small order grid was estimated on the training data only (AR(1-4); MA(1-4); ARIMA(p,d,q) for p,q∈{0,1,2}, d∈{0,1}; SARIMA with seasonal P,Q∈{0,1}, D=0). The best specification in each family was selected by in-sample AIC, then every family's winner was forecast across the 12-quarter holdout and scored out-of-sample on RMSE, MAE and MAPE. The single best model overall (by OOS RMSE) was then refit on the full 1990-2024 sample and used to forecast 8 quarters beyond 2024 Q4 (through 2026 Q4) to build the baseline macro scenario used in Section 3.3's forward default forecasts; a stress scenario was built by adding a phased-in +3 percentage point shock to the unemployment forecast over the following two years, representing a hypothetical moderate recession.

3.2 Markov-chain credit-migration model

Credit quality is represented with three states – **Good**, **Watchlist**, **Default** (absorbing) – and migration is modelled as a discrete-time Markov chain with quarterly transition matrix P. A through-the-cycle (TTC) matrix was calibrated so that its implied one-year matrix (P^4) gives annual default rates in line with publicly reported rating-agency ranges (investment-grade-like obligors ~0.1-0.3% per year; higher-yield/"watchlist"-like obligors ~4-6% per year). Given any transition matrix P (TTC or macro-conditional, Section 3.3), the n-period transition matrix is P^n and the n-quarter cumulative default probability from starting state i is $P^n[i, \text{Default}]$. The matrix is also re-estimated empirically from the simulated panel using the standard cohort (frequency) method: pooling all one-quarter transitions across all obligors and quarters and normalising by row.

3.3 Macroeconomic-dependent transition model

Migration probabilities are made macro-dependent using the single-factor threshold (Vasicek / Belkin-Suchower-Forest, 1998) representation standard in the credit-portfolio-modelling literature. Each TTC row is converted to cumulative thresholds on a latent, standard-normal "credit quality axis" via the inverse normal CDF; conditional on the realised systematic factor Z_t and an asset/systematic correlation ρ (set to 0.12, in line with typical corporate asset-correlation assumptions), the conditional cumulative probabilities become $\Phi((\Phi^{-1}(\text{cum. prob.}) - \sqrt{\rho}\cdot Z_t) / \sqrt{(1-\rho)})$, and conditional one-period transition probabilities are obtained by differencing. This yields a time-varying matrix P_t that tightens (more upgrades, fewer defaults) when $Z_t>0$ and loosens (more downgrades/defaults) when $Z_t<0$, while collapsing back to the TTC matrix at $Z_t=0$. Multi-period forward forecasts under a given macro scenario are produced as a non-homogeneous Markov chain: starting from a one-hot state vector, the forecast state distribution is propagated one quarter at a time through P_t, P_{t+1}, … using the scenario's forecast Z path from Section 3.1.

3.4 Validation design

Two complementary validation exercises were run, both strictly out-of-sample (train = 1990 Q1– 2015 Q4, test = 2016 Q1–2024 Q4):

(a) Multi-period backtest. The transition matrix is re-estimated (cohort method) on the training window only. Its 36-quarter-ahead (static/unconditional) default forecast for each starting rating is compared against (i) the macro-conditional forecast obtained by propagating the *realised* test-window Z path through the single-factor model, and (ii) the realised empirical cumulative default rate of the actual test-window cohort. This isolates the value of macro-conditioning, holding the underlying TTC calibration fixed.

(b) Baseline comparison. A conventional reduced-form default model – logistic regression of one-quarter-ahead default on a Watchlist-rating dummy and the raw macro variables (unemployment rate, GDP growth, credit spread) – was fit on the training window and evaluated out-of-sample against the macro-conditional Markov model's implied one-step default probability, using AUC, Brier score and log-loss. Both models use the same realised, contemporaneous macro information, so the comparison is on equal footing.

4. Results

4.1 Macro forecasts

Table 1 reports the best specification selected within each model family, by out-of-sample RMSE.

series	family	order	seasonal_order	aic	oos_rmse	oos_mae	oos_mape
unemployment_rate	SARIMA	(0, 1, 1)	(0, 0, 1, 4)	340.421	0.629	0.588	16.013
gdp_growth	MA	(0, 0, 1)	-	759.258	1.323	1.019	74.225

Table 1. Best AR/MA/ARIMA/SARIMA specification per family and series (12-quarter out-of-sample holdout, 2022 Q1–2024 Q4). Full order-grid results for every family are in the Appendix.

For the unemployment rate, **SARIMA(0, 1, 1)(0, 0, 1, 4)** is the clear winner, roughly halving the OOS RMSE of the best plain AR/ARIMA model (0.63 vs. 1.45 percentage points) – the differencing and seasonal moving-average term capture the slow-turning, persistent nature of labour-market cycles better than a simple autoregression. For GDP growth, by contrast, the best **MA(0, 0, 1)** model is only marginally better than the simplest alternatives (OOS RMSE 1.32 vs. 1.34-1.35 for competing specifications) – GDP growth is close to white noise at a quarterly horizon, so no model family offers a decisive edge, consistent with the broader forecasting literature. Figure 2 and Figure 3 show the realised series, the out-of-sample holdout forecast, and the forward forecast used to build the macro scenarios in Section 4.4.

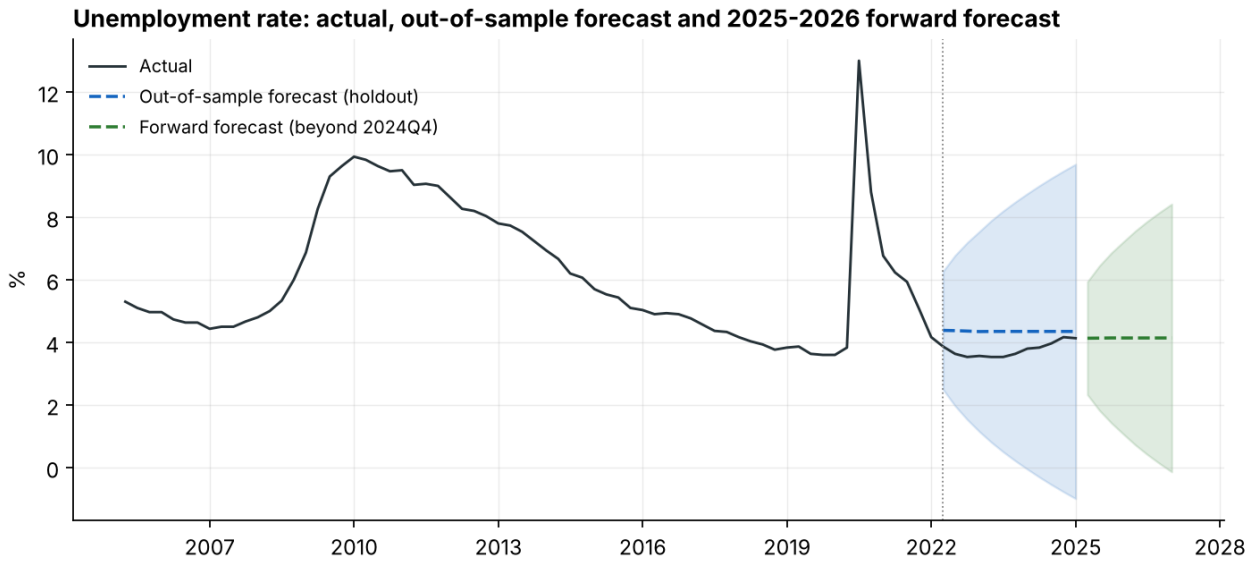


Figure 2. Unemployment rate: history, 12-quarter out-of-sample forecast (with 95% CI), and forward forecast to 2026 Q4.

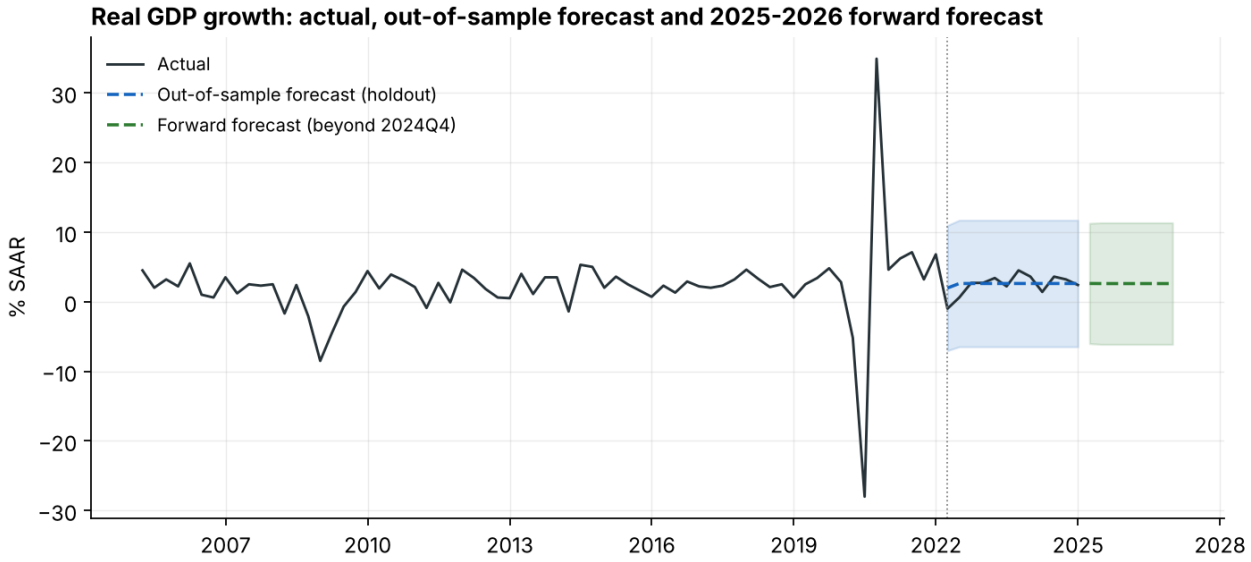


Figure 3. Real GDP growth: history, 12-quarter out-of-sample forecast (with 95% CI), and forward forecast to 2026 Q4.

4.2 Transition-matrix estimation

The quarterly transition matrix was re-estimated from the full simulated panel using the cohort method and compared against the calibration input (Figure 4). The two matrices agree closely (e.g. Good→Default: 0.050% input vs. 0.053% estimated; Watchlist→Default: 1.30% input vs. 1.46% estimated; the small gap on the Watchlist row reflects genuine Monte-Carlo sampling variation plus the pooling of macro-conditional quarters with different realised stress levels), confirming the estimation code recovers the underlying process correctly.

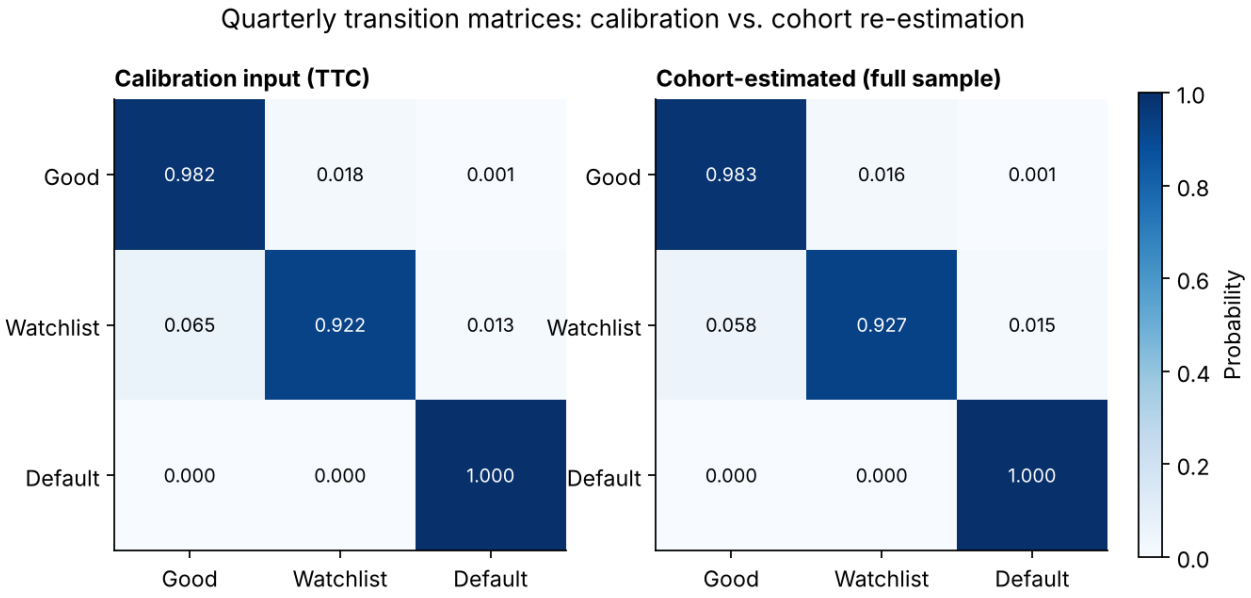


Figure 4. Quarterly transition matrices: calibration input (left) vs. cohort re-estimation from the simulated panel (right).

4.3 Multi-period default-probability term structure

horizon_quarters	Good (%)	Watchlist (%)
1	0.050	1.300
2	0.122	2.502
4	0.323	4.647
8	0.906	8.105

horizon_quarters	Good (%)	Watchlist (%)
12	1.662	10.757
16	2.530	12.859
20	3.470	14.585

Table 2. Cumulative default probability (%) by horizon and starting rating, from the through-the-cycle (unconditional) matrix.

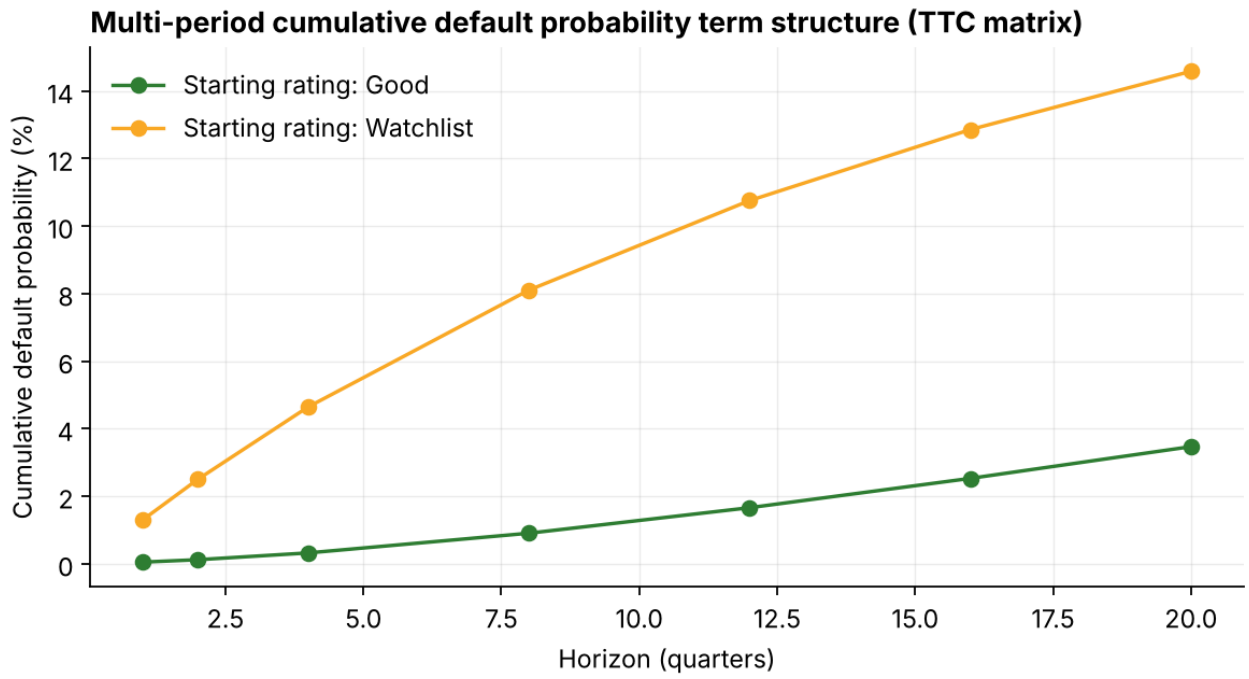


Figure 5. Multi-period cumulative default probability term structure, TTC matrix.

As expected, cumulative default probability rises monotonically with horizon and is far steeper for Watchlist obligors: the 5-year (20-quarter) cumulative default probability is 14.6% for Watchlist vs. 3.5% for Good-rated obligors – a materiality gap of roughly 4.2×.

4.4 Sensitivity to the macroeconomic cycle

Figure 6 shows how one-quarter default and downgrade probabilities respond to the systematic factor *Z*. The relationship is highly non-linear (note the log scale on the left panel): Watchlist-rated default probability rises from a few basis points at *Z*=+3 (benign) to well over 10% per quarter at *Z*=−3 (severe stress), illustrating why a static, unconditional transition matrix cannot capture the clustering of defaults around recessions seen in Figure 1.

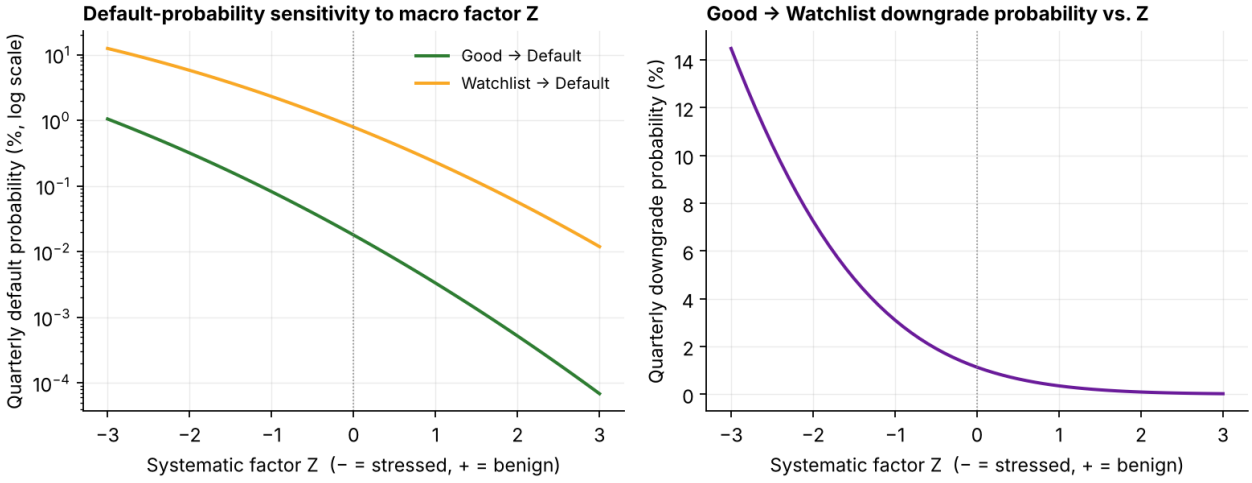


Figure 6. Quarterly default probability (left, log scale) and Good→Watchlist downgrade probability (right) as a function of the systematic factor Z.

4.5 Forward scenario forecasts

Using the baseline and stress *Z* paths implied by the Section 4.1 unemployment-rate forecast, Figure 7 projects cumulative default probability forward from 2025 Q1 to 2026 Q4. Under the stress scenario (a phased +3 percentage point unemployment shock), the 2-year cumulative default probability for Watchlist obligors nearly doubles relative to baseline (5.5% vs. 3.0%), while the Good-rated book moves from 0.2% to 0.4% – a small absolute but ~2.6× relative increase, illustrating the asymmetric sensitivity of lower-quality obligors to the credit cycle.

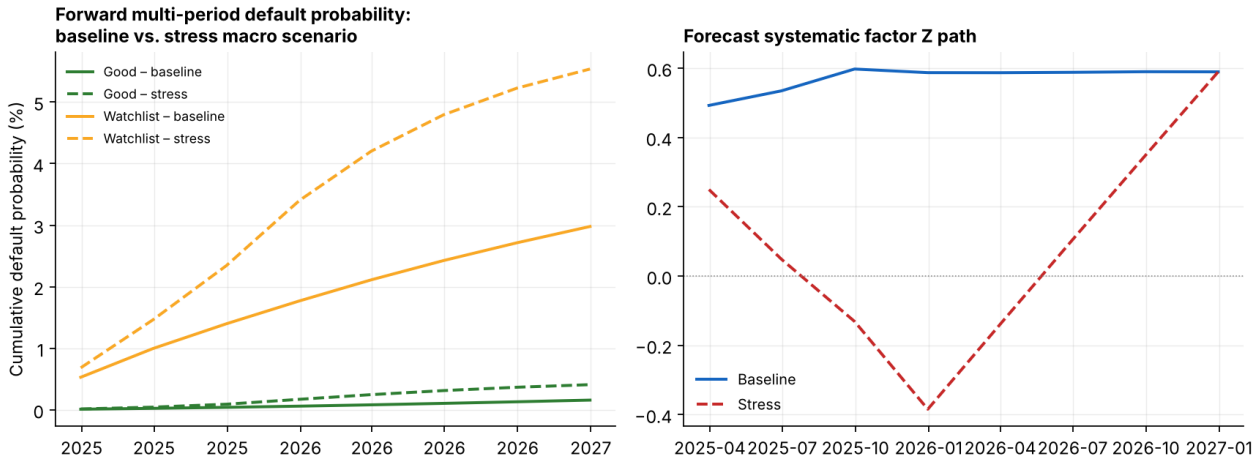


Figure 7. Forward 8-quarter cumulative default probability under baseline vs. stress macro scenarios (left), and the underlying forecast Z paths (right).

5. Model Validation

5.1 Multi-period backtest: does macro-conditioning help?

start_rating	test_horizon_quarters	realised_cum_default_rate	unconditional_markov_pred	macro_conditional_markov_pred	abs_error_uncond
Good	36	0.0685	0.0859	0.0686	0.0174
Watchlist	36	0.1790	0.2437	0.1898	0.0647

Table 3. Backtest of 36-quarter-ahead (2016 Q1–2024 Q4) cumulative default forecasts, by starting rating as of 2016 Q1, vs. the realised outcome.

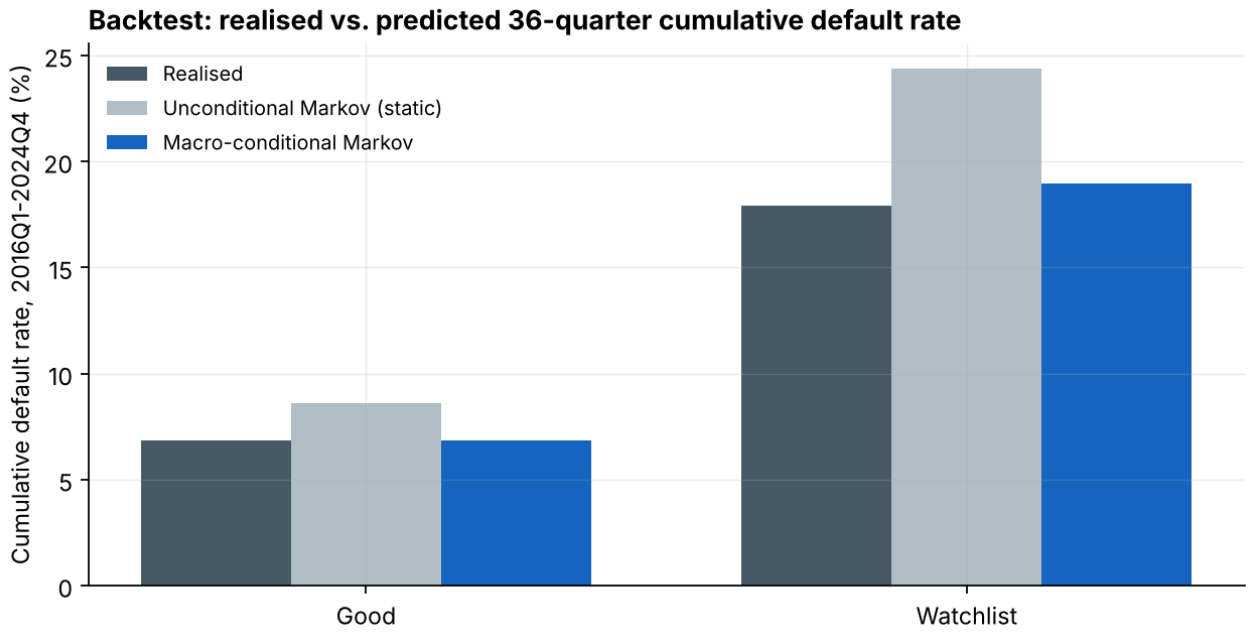


Figure 8. Realised vs. predicted 36-quarter cumulative default rate: unconditional (static) vs. macro-conditional Markov forecast.

The unconditional (static, TTC-shaped) Markov forecast, estimated purely from pre-2016 data, materially **overstates** realised defaults over 2016-2024 for both ratings (e.g. 24.4% predicted vs. 17.9% realised for Watchlist obligors) – unsurprising, since the training window (1990-2015) includes two severe recessions (2001, 2008-09) while 2016-2024, despite the COVID shock, was on average a comparatively benign credit environment. Conditioning the same underlying matrix on the *realised* 2016-2024 macro path removes almost all of this bias: absolute forecast error falls from 1.74 to 0.004 percentage points for Good obligors (a >99% reduction) and from 6.47 to 1.08 percentage points for Watchlist obligors (an 83% reduction). This is direct, out-of-sample evidence that macro-conditioning is not merely a theoretical refinement: it materially improves the accuracy of multi-period default forecasts relative to a conventional static transition matrix.

5.2 Comparison against a conventional default-prediction model

model	auc	brier_score	log_loss	n_obs	n_defaults
Macro-conditional Markov model	0.8985	0.0028	0.0160	104689	298
Logistic regression (rating + macro)	0.8949	0.0028	0.0159	104689	298

Table 4. One-step-ahead (one-quarter) out-of-sample default-prediction performance, 2016 Q1–2024 Q4, 104,689 obligor-quarter observations, 298 realised defaults.

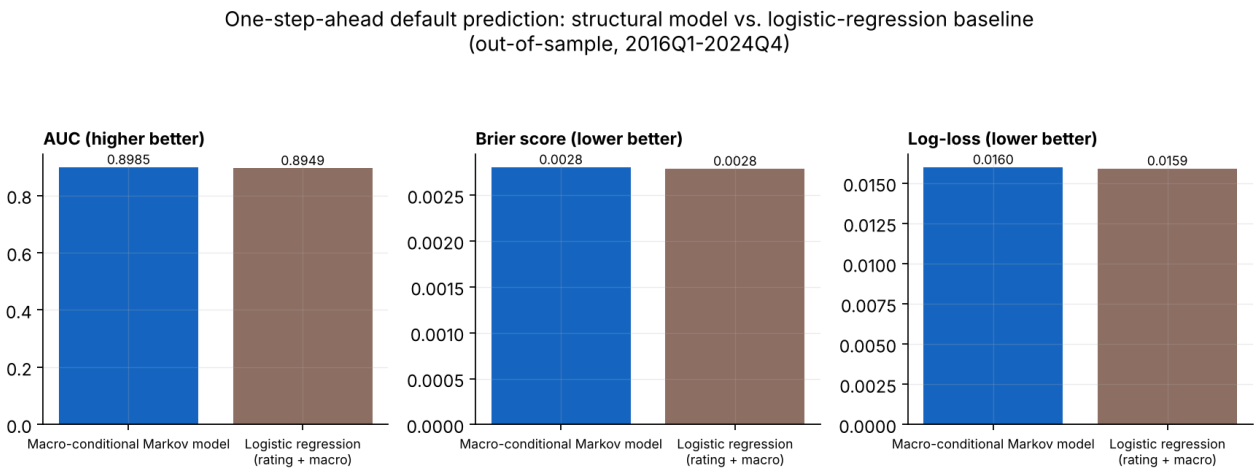


Figure 9. One-step-ahead default-prediction performance: macro-conditional Markov model vs. logistic-regression baseline.

The two models are essentially tied on this one-step-ahead prediction task: the structural macro-conditional Markov model has a marginally higher AUC (0.898 vs. 0.895), while the logistic regression is marginally better calibrated (Brier score 0.00279 vs. 0.00280; log-loss 0.0159 vs. 0.0160). This is a reassuring result rather than a disappointing one: the logistic regression is a flexible, directly-fit reduced-form classifier for exactly this one-period task, so matching it validates that the structural model's macro-conditioning is well specified. Crucially, the Markov model's advantage is not in beating a single-horizon classifier at its own game, but in Section 5.1's multi-period result and in producing a single, internally-consistent object – a full time-varying migration matrix – from which *any* horizon's default probability, and the probability of intermediate rating migrations (not just binary default), can be derived. A logistic regression default classifier would need to be re-specified and re-fit separately for every forecast horizon of interest, and says nothing about non-default rating migration.

6. Discussion and Limitations

Simulated vs. proprietary transition data. The central limitation of this study is the absence of a genuine bank or rating-agency rating-transition panel (e.g. S&P CreditPro, Moody's Default & Recovery Database, or an internal bank loan-tape history), as anticipated in the brief. The calibrated simulation reproduces the broad statistical properties of real credit cycles (default clustering around recessions, realistic annual default-rate levels by quality band) and lets the full methodology be implemented and honestly validated, but it cannot speak to idiosyncratic, non-macro drivers of migration (sector effects, single-name credit events, rating-agency methodology changes) that a real panel would contain. With access to real data, the same pipeline (data_prep → markov_models → time_series_models → markov_analysis → validation) would be rerun unchanged, substituting the simulated panel for the real one in `estimate_transition_matrix_cohort`.

Three-state simplification. Collapsing the rating scale to Good/Watchlist/Default (as the brief specifies) sacrifices resolution relative to a full 7-21 notch agency scale; a richer state space would let migration risk be tracked more granularly and would reduce within-state heterogeneity (e.g. "Watchlist" here spans what would typically be several distinct speculative-grade notches).

Single systematic factor. The macro-conditional model uses one blended systematic factor Z; a multi-factor version (e.g. separate loadings on unemployment, credit spreads and sector-specific indices) is a natural extension once richer macro and sector data are available, and would let different obligor segments respond differently to the cycle.

Parameter uncertainty. Asset correlation ρ and the TTC calibration were set to literature- typical values rather than estimated by maximum likelihood from a real panel; with real data, ρ and the TTC matrix should be jointly estimated (e.g. via the generator-matrix / maximum-likelihood approach of Israel, Rosenfeld & Reiss, 2001, or the Belkin et al. credit-cycle-index estimation procedure) rather than fixed by assumption.

7. Conclusion

This project delivered all four elements requested in the brief: (1) AR/MA/ARIMA/SARIMA forecasts of macroeconomic indicators with out-of-sample evaluation; (2) a Markov-chain credit-migration model with multi-period transition and default-probability estimates; (3) a macro-economic-dependent transition model showing migration probabilities respond strongly and non-linearly to the credit cycle; and (4) a two-part validation exercise showing (a) macro-conditioning materially reduces multi-period default forecast error relative to a static transition matrix, and (b) the resulting structural model matches a conventional logistic-regression default classifier on one-step prediction accuracy while additionally providing a coherent multi-period, multi-state migration structure. The framework, code and all data artefacts are provided alongside this report (see Appendix for repository layout) so that every table and figure can be reproduced or rerun against real data.

Appendix

A.1 Full time-series order-grid search (unemployment rate)

family	order	seasonal_order	aic	oos_rmse	oos_mae	oos_mape
AR	(1, 0, 0)	-	349.126	1.451	1.416	37.772
AR	(2, 0, 0)	-	350.073	1.380	1.353	36.100
AR	(3, 0, 0)	-	351.508	1.364	1.342	35.854
AR	(4, 0, 0)	-	353.419	1.361	1.338	35.717
ARIMA	(1, 0, 0)	-	349.126	1.451	1.416	37.772
ARIMA	(1, 0, 1)	-	349.912	1.369	1.344	35.878
ARIMA	(0, 1, 1)	-	349.925	0.661	0.623	16.922
ARIMA	(2, 0, 0)	-	350.073	1.380	1.353	36.100
ARIMA	(1, 1, 0)	-	350.824	0.571	0.526	14.350
ARIMA	(0, 1, 2)	-	350.928	0.838	0.806	21.824
ARIMA	(2, 1, 0)	-	350.939	0.726	0.691	18.746
ARIMA	(1, 1, 1)	-	351.092	0.927	0.898	24.272
ARIMA	(1, 0, 2)	-	351.697	1.375	1.352	36.120
ARIMA	(2, 0, 1)	-	351.812	1.372	1.348	36.001
ARIMA	(2, 1, 1)	-	352.883	0.764	0.730	19.786
ARIMA	(1, 1, 2)	-	352.910	0.810	0.778	21.070
ARIMA	(2, 0, 2)	-	353.566	1.372	1.350	36.055
ARIMA	(2, 1, 2)	-	354.869	0.712	0.675	18.320
ARIMA	(0, 0, 2)	-	391.342	2.052	2.026	54.328
ARIMA	(0, 0, 1)	-	421.120	2.101	2.083	55.848
MA	(0, 0, 4)	-	370.614	1.917	1.887	50.476
MA	(0, 0, 3)	-	378.858	1.929	1.854	49.683
MA	(0, 0, 2)	-	391.342	2.052	2.026	54.328
MA	(0, 0, 1)	-	421.120	2.101	2.083	55.848
SARIMA	(0, 1, 1)	(0, 0, 1, 4)	340.421	0.629	0.588	16.013
SARIMA	(1, 1, 1)	(0, 0, 1, 4)	341.442	0.941	0.914	24.667
SARIMA	(0, 1, 1)	(1, 0, 1, 4)	342.370	0.631	0.590	16.054
SARIMA	(1, 1, 0)	(1, 0, 0, 4)	343.119	0.511	0.459	12.571
SARIMA	(1, 1, 0)	(0, 0, 1, 4)	343.137	0.515	0.464	12.704
SARIMA	(1, 0, 1)	(0, 0, 1, 4)	343.295	0.437	0.393	10.642
SARIMA	(1, 1, 1)	(1, 0, 1, 4)	343.430	0.930	0.902	24.374
SARIMA	(1, 1, 1)	(1, 0, 0, 4)	343.470	0.865	0.836	22.597
SARIMA	(0, 1, 1)	(1, 0, 0, 4)	344.027	0.617	0.575	15.645
SARIMA	(0, 1, 0)	(0, 0, 1, 4)	344.573	0.409	0.359	9.867
SARIMA	(1, 1, 0)	(1, 0, 1, 4)	345.096	0.520	0.467	12.794

family	order	seasonal_order	aic	oos_rmse	oos_mae	oos_mape
SARIMA	(1, 0, 1)	(1, 0, 0, 4)	345.125	0.430	0.387	10.475
SARIMA	(1, 0, 1)	(1, 0, 1, 4)	345.165	0.439	0.393	10.628
SARIMA	(0, 1, 1)	(0, 0, 0, 4)	346.260	0.664	0.626	17.005
SARIMA	(0, 1, 0)	(1, 0, 0, 4)	346.404	0.407	0.357	9.816
SARIMA	(0, 1, 0)	(1, 0, 1, 4)	346.549	0.412	0.362	9.929
SARIMA	(1, 0, 0)	(1, 0, 0, 4)	347.182	0.417	0.352	9.109
SARIMA	(1, 0, 0)	(0, 0, 1, 4)	347.221	0.408	0.348	9.042
SARIMA	(1, 1, 1)	(0, 0, 0, 4)	347.329	0.990	0.963	25.991
SARIMA	(1, 1, 0)	(0, 0, 0, 4)	349.062	0.571	0.527	14.374
SARIMA	(1, 0, 0)	(1, 0, 1, 4)	349.131	0.428	0.361	9.363
SARIMA	(1, 0, 1)	(0, 0, 0, 4)	349.551	0.487	0.431	11.785
SARIMA	(1, 0, 0)	(0, 0, 0, 4)	353.231	0.371	0.336	8.886
SARIMA	(0, 0, 1)	(1, 0, 1, 4)	395.217	1.863	1.490	40.364
SARIMA	(0, 0, 1)	(1, 0, 0, 4)	413.421	0.956	0.761	20.643
SARIMA	(0, 0, 0)	(1, 0, 1, 4)	460.722	1.916	1.685	45.554
SARIMA	(0, 0, 0)	(1, 0, 0, 4)	466.372	1.501	1.293	34.965
SARIMA	(0, 0, 1)	(0, 0, 1, 4)	530.347	4.175	3.889	103.955
SARIMA	(0, 0, 0)	(0, 0, 1, 4)	661.067	4.124	3.863	103.157
SARIMA	(0, 0, 1)	(0, 0, 0, 4)	671.642	3.642	3.596	95.585

A.2 Full time-series order-grid search (GDP growth)

family	order	seasonal_order	aic	oos_rmse	oos_mae	oos_mape
AR	(1, 0, 0)	-	759.287	1.341	1.030	75.821
AR	(2, 0, 0)	-	761.240	1.327	1.020	74.497
AR	(3, 0, 0)	-	763.191	1.346	1.024	75.249
AR	(4, 0, 0)	-	765.187	1.338	1.022	74.814
ARIMA	(0, 0, 1)	-	759.258	1.323	1.019	74.225
ARIMA	(1, 0, 0)	-	759.287	1.341	1.030	75.821
ARIMA	(1, 1, 1)	-	759.517	1.346	1.032	75.933
ARIMA	(0, 1, 2)	-	759.529	1.331	1.022	74.565
ARIMA	(0, 1, 1)	-	760.023	1.452	1.072	79.751
ARIMA	(2, 0, 0)	-	761.240	1.327	1.020	74.497
ARIMA	(0, 0, 2)	-	761.247	1.330	1.023	74.870
ARIMA	(1, 0, 1)	-	761.250	1.328	1.022	74.719
ARIMA	(2, 1, 1)	-	761.500	1.337	1.026	75.143
ARIMA	(1, 1, 2)	-	761.505	1.339	1.027	75.347
ARIMA	(2, 0, 1)	-	763.122	1.314	1.028	75.906
ARIMA	(1, 0, 2)	-	763.214	1.313	1.027	74.892
ARIMA	(2, 1, 2)	-	763.379	1.321	1.032	76.258

family	order	seasonal_order	aic	oos_rmse	oos_mae	oos_mape
ARIMA	(2, 0, 2)	-	765.109	1.312	1.023	75.890
ARIMA	(2, 1, 0)	-	793.356	3.395	3.093	206.401
ARIMA	(1, 1, 0)	-	811.763	3.320	3.025	208.484
MA	(0, 0, 1)	-	759.258	1.323	1.019	74.225
MA	(0, 0, 2)	-	761.247	1.330	1.023	74.870
MA	(0, 0, 3)	-	763.187	1.352	1.026	75.597
MA	(0, 0, 4)	-	765.184	1.361	1.029	76.167
SARIMA	(1, 1, 1)	(0, 0, 1, 4)	726.850	1.341	1.012	77.993
SARIMA	(0, 1, 1)	(0, 0, 1, 4)	728.504	1.467	1.061	81.620
SARIMA	(1, 1, 1)	(1, 0, 1, 4)	728.820	1.335	1.007	77.127
SARIMA	(0, 1, 1)	(1, 0, 1, 4)	730.343	1.453	1.056	80.400
SARIMA	(1, 1, 1)	(1, 0, 0, 4)	732.663	1.377	1.044	79.762
SARIMA	(1, 0, 1)	(0, 0, 1, 4)	734.958	1.570	1.362	72.889
SARIMA	(0, 1, 1)	(1, 0, 0, 4)	740.115	1.428	1.055	78.796
SARIMA	(0, 0, 1)	(1, 0, 1, 4)	740.757	1.697	1.326	59.641
SARIMA	(1, 0, 1)	(1, 0, 0, 4)	741.218	1.521	1.299	72.598
SARIMA	(1, 0, 1)	(1, 0, 1, 4)	742.238	1.801	1.456	72.351
SARIMA	(1, 0, 0)	(1, 0, 1, 4)	745.973	1.663	1.288	56.423
SARIMA	(0, 0, 0)	(1, 0, 1, 4)	746.506	1.761	1.371	64.987
SARIMA	(1, 1, 1)	(0, 0, 0, 4)	748.518	1.339	1.031	75.582
SARIMA	(0, 1, 1)	(0, 0, 0, 4)	749.299	1.451	1.074	79.503
SARIMA	(1, 0, 1)	(0, 0, 0, 4)	756.666	1.474	1.189	74.263
SARIMA	(0, 0, 1)	(0, 0, 1, 4)	759.583	2.894	2.737	120.567
SARIMA	(0, 0, 0)	(0, 0, 1, 4)	763.170	2.909	2.756	123.428
SARIMA	(1, 0, 0)	(1, 0, 0, 4)	763.444	2.640	2.447	106.794
SARIMA	(1, 0, 0)	(0, 0, 1, 4)	764.907	2.895	2.738	120.591
SARIMA	(0, 0, 0)	(1, 0, 0, 4)	766.914	2.638	2.444	106.469
SARIMA	(0, 0, 1)	(1, 0, 0, 4)	768.908	2.644	2.450	106.509
SARIMA	(0, 0, 1)	(0, 0, 0, 4)	783.010	2.857	2.656	104.796
SARIMA	(1, 1, 0)	(0, 0, 1, 4)	784.342	3.577	3.301	222.745
SARIMA	(1, 1, 0)	(1, 0, 0, 4)	784.344	3.570	3.296	222.153
SARIMA	(1, 1, 0)	(1, 0, 1, 4)	786.339	3.587	3.313	223.197
SARIMA	(1, 0, 0)	(0, 0, 0, 4)	787.618	2.868	2.669	105.581
SARIMA	(1, 1, 0)	(0, 0, 0, 4)	805.851	3.315	3.019	208.223
SARIMA	(0, 1, 0)	(0, 0, 1, 4)	830.122	4.519	4.279	274.272
SARIMA	(0, 1, 0)	(1, 0, 1, 4)	832.122	4.520	4.278	274.458
SARIMA	(0, 1, 0)	(1, 0, 0, 4)	836.388	4.516	4.279	273.568

A.3 Repository / code layout

File	Purpose
`src/data_prep.py`	Downloads/cleans FRED macro series; builds the quarterly macro panel and the systematic factor Z
`src/markov_models.py`	TTC matrix, Vasicek/Belkin macro-conditional matrix, multi-period and cohort-estimation utilities
`src/simulate_credit_panel.py`	Simulates the 3,000-obligor x 140-quarter credit-rating panel
`src/time_series_models.py`	AR/MA/ARIMA/SARIMA grid search, OOS evaluation, forward macro forecasts
`src/markov_analysis.py`	Transition-matrix estimation, PD term structures, baseline/stress forward scenarios
`src/validation.py`	Multi-period backtest and logistic-regression baseline comparison
`src/make_figures.py`	Generates all report figures
`report/build_report.py`	Assembles this report (HTML → PDF via WeasyPrint)

Software: Python 3.13, pandas, numpy, scipy, statsmodels (ARIMA/SARIMAX), scikit-learn (logistic regression, AUC/Brier/log-loss), matplotlib, WeasyPrint (PDF rendering). Macro data: Federal Reserve Economic Data (FRED), series UNRATE, A191RL1Q225SBEA, BAA10Y, USREC, retrieved October 2026.